

- 2 When a barley seed germinates, gibberellin is released from structure **A**. Gibberellin diffuses to its target cells located in tissue **B**. This causes the cells at **B** to synthesise the enzyme amylase.

(a) Figure 2.1 is a diagram of a longitudinal section through a barley seed.

On Figure 2.1, use label lines and the letters **A** and **B** to identify:

A – the structure where gibberellin is released

B – the location of the target cells for gibberellin that synthesise amylase.

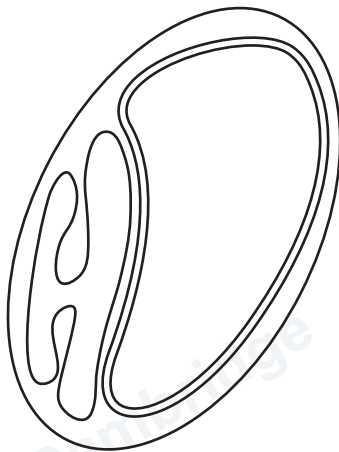


Figure 2.1

[2]

- (b) A gene in the cells of **B** is switched on so that the cells can synthesise amylase.

Figure 2.2 shows the sequence of molecular events that occurs in the cells of **B** during seed germination.

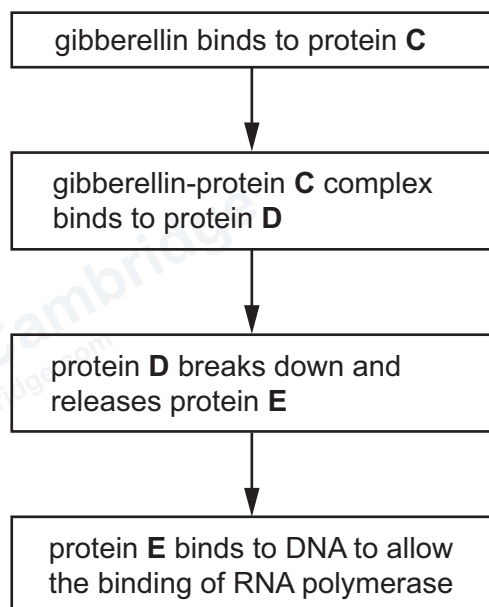


Figure 2.2

- (i) Identify proteins **C**, **D** and **E**.

C

D

E

[3]

- (ii) After protein **E** binds to DNA, RNA polymerase can bind to DNA.

Identify:

- the region of the DNA that RNA polymerase binds to
- the gene product.

region of DNA

gene product

[2]



- (c) A variety of pea plant has a homozygous recessive genotype that prevents it from making active gibberellin.

Suggest what effect this homozygous recessive genotype will have on the height of these pea plants.

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..... [1]

[Total: 8]